Helwan International Technological University

Technological Faculty in Cairo

Program: Industrial control Technology

Level: One

Subject: Mathematics 1

Sheet 1

1. Write each of the following products with a single base:

a. $x^5.x^{-3}.x^2$ b. $2^2.2^6.2$ c. $\frac{x^4}{x^3}$ d. $\frac{5^6}{5^4}$ e. $((2x)^3)^2$

2. Simplify each expression

a. $\frac{2y^6}{y.\ y^5}$ b. $(x^2\sqrt{2})^4(x^2\sqrt{2})^{-4}$ c. $(-2a^3b^{-1})(5a^{-2}b^2)$ d. $\frac{a^5}{a^2.a^4}$

3. Simplify the radical expression

a. $\sqrt{16\ x^4}$ b. $\frac{\sqrt{50x^6y^4}}{\sqrt{2x^2y^4}}$ c. $\sqrt{\frac{36}{25}}$

4. For the following polynomials, identify the degree, the leading term, and the leading coefficient

a. $4x^2 - x^6 + 2x - 6$ b. $4x^3 + 8x^2 - 5x + 20$ c. $14m^3 + m^2 - 16m + 8$

5. Expand the binomial

a. $(3y - 7)^2$ b. $(9b + 1)^2$ c. $(4x + 5)^2$

6. Factor.

a. $x^2 - 7x + 6$. b. $x^2 + 2x - 3$ c. $x^2 - 16$

Choose the correct answer for the following question:

7	$\sqrt{\frac{32x^3}{2x}}$ = ……			
	$3x^2$	4x	$\sqrt{16x}$	$4x^2$
8	**$(x-1)^2$ = ……..**			
	$2x^2$-x-1	(x-1)(x+1)	x^2+2x-1	x^2-2x+1
9	$\left(\frac{a^2}{a^3}\right)^2$ = ……..			
	a^2	a^4	a^{-2}	a
10	$\frac{6c^2c^0}{2c}$ = ……..			
	3c	c	$3c^2$	0
11	**(x+2)(x-2) = …….**			
	$x^2 + 4$	$x^2 - 4$	x + 2	x + 4
12	**x^2 -3x -10 = ……**			
	(x+2)(x-5)	(x-2)(x+5)	(x+2)(x+5)	(x-2)(x-5)
13	**$(a - 4)^2$ = …….**			
	$a^2 + 8a + 16$	a -4a + 16	$a^2 - 4a + 8$	$a^2 - 8a + 16$
14	**$16x^2 – 81$ = …….**			
	(x-9)(x+9)	(4x-3)(4x+3)	(4x-9)(4x+9)	(x-81)(x+81)
15	$\frac{(xy)^4}{y^3} \cdot \frac{2}{x^4}$ = …..			
	2 y	2 xy	xy^2	0